Learning to Fit Shapes to Neural Geometry: A Synthetic-Only Shape-Type and Parameter Regressor for Top-3 PCA Activation Clouds

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Abstract

Interpretability research repeatedly reports low-dimensional shapes in neural activations: circles for weekdays and months, helices for positions and counts, a paraboloid for colors, lines for temperature and age, polytopes for features in superposition. These shapes are usually identified and fitted by hand. We (i) compile a cited catalog of empirically reported neural geometries, (ii) train a permutation-invariant network *only on synthetic shapes* that maps a 3D point cloud (the top-3 principal components of activations) to one of 13 shape families and to a canonical, unique parameter vector, using a shape-to-shape Chamfer loss against the ground-truth shape plus a point-to-shape residual, and (iii) evaluate it on a preregistered synthetic test set and on real activations of GPT-2, Llama-3.1-8B, a grokked modular-addition transformer and toy models of superposition, against a classical fit-every-family baseline with geometric-AIC model selection, a feature-based classifier, trivial baselines, random controls and inference-time ablations. On 39000 held-out synthetic clouds the fitter identifies the family in 87.9% of cases; on a shared subset it reaches 87.4% against 52.2% for the classical pipeline and 64.5% for the feature classifier, with a lower median Chamfer distance to the ground-truth shape (preregistered H1 and H2 supported). On real activations its coarse shape class agrees with published descriptions on 7/12 preregistered datasets, which is not above a permutation null (H3 not supported). Post-hoc inspection points to curves outside the vocabulary, low-variance projections and clouds unlike any training family. Controls show that smooth ordinal processes are often labelled helix, and a preregistered held-out test against the PCA ringing artifact, while necessary, still accepts helix fits on clouds the literature describes as circles.

1 Introduction

A growing body of work shows that some concepts in language models are represented not by single directions but by low-dimensional manifolds [Engels et al., 2025, Bhalla et al., 2026, Gurnee et al., 2025]. The common workflow is: collect activations for a set of inputs that vary along one concept, project onto the top principal components, look at the 2D/3D scatter, and describe the shape ("a circle", "a helix"), often followed by a hand-designed parametric fit [Kantamneni and Tegmark, 2025]. This is slow, subjective, and vulnerable to a known artifact: when the similarity between points decays smoothly with an ordinal or latent variable (distance-decaying covariance, random walks, autocorrelated signals), principal component analysis (PCA) yields Fourier-like components and therefore an arch or a helix-like curve in the top components even when nothing "curved" is represented [Novembre and Stephens, 2008, Diaconis et al., 2008, Antognini and Sohl-Dickstein, 2018, Shinn, 2023].

We ask whether the description step can be automated: given the top-3 PCA coordinates of a set of activations, predict which shape family it is and the parameters that uniquely specify the shape, such that the shape passes as close as possible to the points. Our contributions:

1. A catalog of empirically reported neural geometries with model, layer, method, whether the shape is visible in the top-3 PCs, and verified sources (Sec. 2).

2. A 13-family shape vocabulary covering the catalog entries that can be represented in 3D, with differentiable samplers, label margins between nested families, and canonical forms that make the output parameters unique (Sec. 3), including a family for the PCA "ringing" confounder.

3. A Set Transformer fitter trained only on on-the-fly synthetic shapes with domain randomization, a shape-to-shape Chamfer loss, and test-time refinement (Sec. 4).

4. A preregistered evaluation (hypotheses H1–H5; H5a needs the postponed trained ablations) on synthetic and real activations with baselines, controls, inference-time ablations, bootstrap confidence intervals and multiple-testing correction (Sec. 5–6); both positive and negative results are reported.

2 Background and related work

Empirical neural geometries. Circles for days of the week and months were found with sparse-autoencoder feature clusters in GPT-2 and Mistral-7B, and with PCA of hidden states on day and month arithmetic prompts in Mistral-7B and Llama-3-8B [Engels et al., 2025]; GPT-2's learned positional embeddings form a helix in their top-3 PCs [Yedidia, 2023]; the character count since the last line break lives on a rippled helix-like curve in Claude 3.5 Haiku [Gurnee et al., 2025], a finding reproduced as a helix-like curve in the top-3 PCs of Llama-3.1-8B and Qwen3-8B, with the clearest layers early-to-mid (3–6) rather than very early [Sinii and Balagansky, 2026]; numbers form a generalized helix [Kantamneni and Tegmark, 2025]; Llama-3.1-8B layer 19 shows a paraboloid for colors, a helix for years, a circle for days and lines for temperature and age [Bhalla et al., 2026]; in grokked modular-addition transformers the token embeddings lie on circles, one per key frequency [Nanda et al., 2023, Zhong et al., 2023], and in the "pizza" algorithm the averaged embeddings of input pairs fill the disc inside the circle [Zhong et al., 2023]; features in superposition form uniform polytopes such as antipodal pairs, triangles, pentagons, tetrahedra and square antiprisms [Elhage et al., 2022]; true and false statements separate in the top two principal components [Marks and Tegmark, 2024]; supervised MDS recovers circles, lines, clusters and, for cities, a sphere [Tiblias et al., 2026]; the sky positions of stars and constellations form a sphere, though not within the top three PCs [Berdnikov and Liokumovich, 2026]; in-context graphs reorganize into rings and grids [Park et al., 2025a]; emotion steering vectors arrange approximately on a circle [Sun et al., 2026]. Outside LLMs, grid cells form a torus [Gardner et al., 2022], head-direction cells a ring (visualised with Isomap, as PCA often fails to show it; Chaudhuri et al., 2019), natural image patches a Klein bottle [Carlsson et al., 2008]. Appendix A (generated from `catalog.yaml`) lists all entries.

Primitive fitting. Deep primitive fitting on CAD point clouds predicts per-point memberships and fits planes, spheres, cylinders and cones with differentiable least squares [Li et al., 2019], B-spline patches [Sharma et al., 2020] or regressed parameters [Yan et al., 2021]; the classical alternative is RANSAC [Schnabel et al., 2007] with information-criterion model selection [Kanatani, 2004]. These methods target large CAD clouds with several primitives; neural-geometry clouds have one shape, few points (7 weekdays), high noise, families absent from CAD vocabularies (helix, torus, PCA ringing), and come from a PCA projection of a much higher-dimensional space. To our knowledge, no prior work learns shape-type classification plus parameter regression for activation point clouds. The closest are hypothesis-driven supervised MDS, which needs concept labels and has no free shape parameters [Tiblias et al., 2026], and manifold fitting with cubic or thin-plate splines through concept centroids, whose shape (closed loop, open curve, sheet) is chosen by hand from the known task structure [Wurgaft et al., 2026].

3 Shape vocabulary

Each family f maps unconstrained raw parameters θ to a shape $S_f(\theta) \subset \mathbb{R}^3$, posed by a center c and a rotation $R \in SO(3)$ predicted with the continuous 6D representation [Zhou et al., 2019]. Table 1 lists the families with their parameters. Each family provides (i) a differentiable dense sampler on a fixed grid with length/area weights, (ii) label margins so that a generated instance is not (nearly) a member of a simpler family at the sample's noise level (e.g. an arc's sagitta must exceed $\max(0.1\,\text{chord}, 3\sigma)$), and (iii) a canonicalization map that selects a unique representative among parameterizations of the same point set (axis signs fixed by the sign of the largest component; rotation about symmetry axes fixed by projecting the first PCA axis, or the second when the first is nearly parallel to the symmetry axis; point sets sorted by their first PCA coordinate, with approximate tie-breaks on the second and third). The *ringing* family $c + \sum_{k=1}^{3} a_k \cos(k\pi s) e_k$ is the shape PCA produces when similarity decays smoothly with an ordinal variable [Novembre and Stephens, 2008, Antognini and Sohl-Dickstein, 2018], so a helix claim must beat it. The *polytope* family has six templates: the regular

Table 1: The 13 shape families. Local shapes are posed by a center c and a frame $R = (e_1, e_2, e_3)$ unless stated. Canonical forms make the reported parameters unique.

Family	Local shape	Canonical unique parameters	Anchor
blob	$\mathrm{diag}(\sigma_1, \sigma_2, \sigma_3)z$, $z \sim \mathcal{N}(0, I)$	center, axes (signs fixed), $\sigma_1 \geq \sigma_2 \geq \sigma_3$	null class
clusters	$k \in \{2..6\}$ centroids + spread	centroids sorted by PC1, spread	true/false separation [Marks and Tegmark, 2024]
polytope	template: 5 regular (triangle, square, pentagon, tetrahedron, octahedron) + square antiprism	subtype, center, circumradius, vertices sorted by PC1	superposition [Elhage et al., 2022]
line	tLe_1, $t \in [-1, 1]$	endpoints	temperature, age [Bhalla et al., 2026]
arc	$r(\cos t\, e_1 + \sin t\, e_2)$, $\lvert t \rvert \leq \Delta/2$	center, normal, r, extent (closed if $\Delta \approx 2\pi$)	days, months [Engels et al., 2025]
ellipse	$a \cos t\, e_1 + b \sin t\, e_2$	center, normal, major axis, $a \geq b$	distorted circle
helix	$r \cos t\, e_1 + r \sin t\, e_2 + \frac{pt}{2\pi} e_3$	axis, r, signed pitch p, turns, phase	GPT-2 W_{pos} [Yedidia, 2023]
ringing	$\sum_{k=1}^{3} a_k \cos(k\pi s) e_k$	$a_1, a_2 > 0$, a_3, frame	PCA artifact [Novembre and Stephens, 2008]
disc	$\rho(a \cos t\, e_1 + b \sin t\, e_2)$	center, normal, $a \geq b$	pizza (pair embeddings) [Zhong et al., 2023]
sphere	cap of half-angle α around e_3	center, r, pole, α	cities [Tiblias et al., 2026], sky [Berdnikov and Liokumovich, 2026]
paraboloid	$z = \frac{1}{2}(\kappa_1 x^2 + \kappa_2 y^2)$, $\lvert \kappa_2 \rvert \leq \kappa_1$	frame, κ_1, κ_2, patch radius	colors [Bhalla et al., 2026]
cone	frustum $r(h) = r_0 + gh$ around e_3	axis, bottom/top radius, height	"narrow cone" anisotropy [Ethayarajh, 2019]
torus	$((R + r\cos v)\cos u, (R + r\cos v)\sin u, r \sin v)$	center, axis, R, r	grid cells [Gardner et al., 2022]

triangle, square, pentagon, tetrahedron and octahedron, and the square antiprism, which is uniform but not regular.

4 Method

Preprocessing. Synthetic and real clouds pass through the same map: center, project onto the top-3 principal directions, fix axis signs by the sign of the third moment, force a right-handed frame (3D inputs), and divide by the RMS radius. No whitening is applied, so aspect ratios are preserved.

Synthetic data. Each training step draws fresh shapes for all 13 families from a dedicated seed stream (test and validation streams are disjoint by construction). Nuisances implement domain randomization [Tobin et al., 2017]: $N \in [5, 1024]$ log-uniform, relative noise $\sigma \in [0.003, 0.15]$ log-uniform (10% noise-free), anisotropic noise (30% of samples), non-uniform sampling densities along curves and surfaces, 0–10% outliers in 30% of samples, and, with probability 0.25 per family batch, embedding into $\mathbb{R}^D$ ($D \in \{16, 64, 256\}$) with noise in all D dimensions followed by PCA back to 3D, which reproduces the real pipeline including projection effects and mirrored frames.

Model. A Set Transformer encoder [Lee et al., 2019] (two ISAB blocks with 16 inducing points, PMA pooling, $d = 128$, 4 heads, key-padding masks) maps per-point features (xyz and $\log N$) to an embedding; a classification head outputs logits over the 13 families and one MLP head per family outputs its raw parameters.

Loss. For a sample of family y with ground-truth shape S^* and input points X,

$$\mathcal{L} = \mathrm{CE}(\ell, y) + \mathrm{CD}\big(S_y(\hat{\theta}_y), S^*\big) + 0.5\,\mathrm{Res}\big(X \to S_y(\hat{\theta}_y)\big) + 0.1\,\mathrm{CE}_{\mathrm{sub}} + \lambda(t)\,\mathrm{Aux}, \tag{1}$$

where CD is the symmetric, length/area-weighted, unsquared Chamfer distance [Fan et al., 2017] between dense samples of the predicted and of the noise-free ground-truth shape (restricted to its true extent), Res is a Huber ($\delta = 0.05$) point-to-shape residual, $\mathrm{CE}_{\mathrm{sub}}$ classifies the number of clusters and the polytope template, and Aux is an annealed axis/scale term $1 - (\hat{a} \cdot a^*)^2 + \lvert \log \hat{s} - \log s^* \rvert$ whose weight falls linearly from 1 to 0.1 over the first 30% of the planned steps. The geometric terms are averaged within each family and then over families. Because CD compares point sets it is invariant to every parameter symmetry, so no canonicalization or matching is needed during training. Only the ground-truth family's head receives geometric losses (teacher forcing; SPFN likewise computes its fitting loss with the ground-truth primitive type, Li et al., 2019). Training uses AdamW (learning rate 3×10^{-4}, weight decay 10^{-4}), label smoothing 0.05, a 2k-step warmup and a cosine schedule; the 110-minute wall-clock budget stopped training before the end of the schedule (steps in the models table, Appendix).

Inference. The top-3 families are each refined for 300 Adam steps on the two-sided objective $\frac{1}{N}\sum_i d(x_i, S) + 0.25\sum_j w_j d(s_j, X)$, initialized by the network (amortized inference followed by optimization, cf. Kolotouros et al., 2019); the second term prevents degenerate inflation of radii and extents. The family is selected by the temperature-calibrated classifier [Guo et al., 2017]; geometric AIC [Kanatani, 2004] and a hybrid rule are ablations. The output is mapped back to PCA coordinates and canonicalized.

5 Experimental setup

Preregistration. Hypotheses, metrics, analysis plan and falsification criteria were committed before training (`reports/preregistration.md`); deviations are logged in `reports/deviations.md`.

Synthetic test set. 250 samples per cell of 13 families × 4 point-count bins × 3 noise bins (39,000 clouds) from the `test` stream. Baselines run on a fixed subset of 20 samples per family and cell; all comparisons with baselines use that subset, while the refinement comparison (H5b) is paired on all 39000 clouds.

Baselines. B1: classical multi-start fit of every family with the same two-sided objective (initial frames: the identity, two axis permutations and a random rotation; k-means for clusters; every polytope template), up to 61 refinement runs that leave up to 22 candidates per cloud (cluster counts k not smaller than the smallest N in a batch of 256 clouds are skipped, so in the smallest point-count bin B1 cannot output 5 or 6 clusters), and geometric-AIC selection. B2: hand-crafted features (global PCA eigenvalue ratios; mean and spread over local neighbourhoods of the eigenvalue-based linearity, planarity and scattering [Weinmann et al., 2014, 2015]; persistent-homology H0/H1 lifetimes via ripser [Tralie et al., 2018]; radial and nearest-neighbour statistics) with gradient-boosted trees, then B1's lowest-objective candidate of the predicted family. B3: always blob, always arc (B1's arc fit; the preregistration specified a full circle), random shape.

Ablations. A1 parameter-space loss (its configuration also drops the residual term), A2 no residual term, A3 no refinement, A4 DeepSets encoder [Zaheer et al., 2017], A5 selection rule, A6 no $\mathbb{R}^D$ augmentation; A1/A2/A4/A6 are compared with A0, the unchanged method trained with the same shorter budget. Only the inference-time ablations A3 and A5 are reported here; the trained ones are postponed.

Real activations. R1 GPT-2 positional embeddings, rows 1–1023 [Radford et al., 2019, Yedidia, 2023]; R2 embeddings of a 1-layer modular-addition transformer with the architecture and optimizer of Nanda et al. [2023], trained for 25,000 instead of 40,000 epochs; R3 toy models of superposition with 3 hidden dimensions and 6 or 8 features [Elhage et al., 2022]; R4 Llama-3.1-8B [Grattafiori et al., 2024] layer-5 mean activations per character count since the last newline [Gurnee et al., 2025, Sinii and Balagansky, 2026]; R5 weekday and month tokens in the arithmetic prompts of Engels et al. [2025] (layer 5; our weekday prompt reads "days of the week math" where theirs reads "day of the week math"); R6–R8 years, colors, temperature and age with the templates of Bhalla et al. [2026] (layer 19, last token); R9 true/false city statements [Marks and Tegmark, 2024] (layer 12). "Layer L" is the Hugging Face hidden state after L blocks. Exploratory additions, not part of H3: Pythia-410m [Biderman et al., 2023] counting, a days-and-times prompt, a 20-feature toy model and the class means of R5. Controls: C1 smooth Gaussian processes and random walks over an ordinal index in $\mathbb{R}^{768}$ (expected: ringing); C2 matched-covariance Gaussians for each dataset (expected: blob); C3 the same inputs through randomly initialized weights (for GPT-2 W_{pos}, Hugging Face's $\mathcal{N}(0, 0.02^2)$ initialization).

Statistics. Percentile bootstrap CIs (2000 replicates) and paired bootstrap CIs for every comparison. The p-value of H1 (macro-F1) is a normal approximation from the paired-bootstrap standard error; H2 and H5b use Wilcoxon signed-rank tests on per-cloud Chamfer distances; all are Holm-corrected [Holm, 1979], and values below 10^{-16} are shown as a floor. For real data, 200 random 80% subsamples (PCA recomputed) give label frequencies, and H3 uses the majority coarse label over them; a 5-fold held-out point residual compares method, B1, helix and ringing fits; exact (Clopper–Pearson) binomial CIs and a one-sided label-permutation null (10,000 permutations) test agreement with the literature.

6 Results

6.1 Synthetic test set

Table 2 compares all systems. On the full test set ($n = 39000$ clouds), the method selects the correct family for 87.9% [87.6, 88.2] of the clouds, with a median Chamfer distance to the noise-free ground-truth shape of 0.156 [0.154, 0.158] (in units of the RMS radius of the cloud). On the baseline subset, where every system sees the same $n = 3120$ clouds, the method reaches 87.4%, against 64.5% for the feature classifier B2, 52.2% for the classical fit-every-family pipeline B1, and 7.7%–7.9% for the trivial baselines.

Table 2: Synthetic test set. Fine accuracy is the exact family; coarse accuracy is the shape dimension (point set, curve, surface, blob). Chamfer is the median symmetric Chamfer distance between the selected, refined shape and the noise-free ground-truth shape. Baselines and the "baseline subset" row use the same 20 clouds per family and cell. 95% percentile-bootstrap CIs. The full table (superset-tolerant accuracy, residual, axis error) is in `reports/exp01_synthetic.md`.

System	n	Fine accuracy [95% CI]	Macro-F1 [95% CI]	Coarse accuracy	Median Chamfer to GT [95% CI]
Method (network + refinement)	39000	87.9% [87.6, 88.2]	0.879 [0.876, 0.882]	92.7%	0.156 [0.154, 0.158]
Method, baseline subset	3120	87.4% [86.2, 88.5]	0.873 [0.862, 0.884]	92.3%	0.154 [0.147, 0.161]
A3: no refinement	39000	87.9% [87.6, 88.2]	0.879 [0.876, 0.882]	92.7%	0.159 [0.157, 0.162]
A5: G-AIC selection	39000	64.6% [64.1, 65.1]	0.627 [0.623, 0.631]	77.5%	0.182 [0.179, 0.185]
A5: hybrid selection	39000	85.6% [85.2, 85.9]	0.858 [0.854, 0.861]	91.1%	0.157 [0.155, 0.159]
B1: classical fit + G-AIC	3120	52.2% [50.6, 54.0]	0.499 [0.484, 0.514]	70.7%	0.209 [0.200, 0.221]
B2: features + boosting	3120	64.5% [62.8, 66.2]	0.645 [0.628, 0.661]	85.1%	0.237 [0.221, 0.255]
B3a: always blob	3120	7.7% [6.8, 8.7]	0.011 [0.010, 0.012]	7.7%	0.543 [0.536, 0.550]
B3b: always arc (B1 arc fit)	3120	7.7% [6.8, 8.6]	0.011 [0.010, 0.012]	38.5%	0.545 [0.536, 0.555]
B3c: random shape	3120	7.9% [7.1, 8.8]	0.079 [0.070, 0.088]	32.9%	1.116 [1.100, 1.137]

Preregistered hypotheses (Table 3). H1 is supported: the paired macro-F1 difference between the method and B1 is 0.374 [0.357, 0.391]. H2 is supported: the paired difference in median Chamfer distance is -0.055 [-0.065, -0.046]. Refinement (H5b) lowers the median Chamfer distance by a small but significant amount, -0.003 [-0.005, -0.002]; because the family is selected by the classifier before refinement, refinement cannot change accuracy (A3 has the same accuracy as the method by construction). H5a (Chamfer versus parameter-space loss) requires the ablation runs, which are postponed (Sec. 7).

Table 3: Preregistered hypotheses. p-values are Holm-corrected over the synthetic comparisons (H1: normal approximation from the paired-bootstrap standard error; H2, H5b: Wilcoxon signed-rank); "< 1e-16" is a display floor. H3 uses the subsample-majority coarse label, an exact binomial CI and a 10,000-permutation null over predicted coarse labels.

Hypothesis	Comparison	Estimate [95% CI]	p (Holm)	n	Verdict
H1	macro-F1(method) - macro-F1(B1)	0.3741 [0.3572, 0.3905]	< 1e-16	3120	supported
H2	median Chamfer(method) - median Chamfer(B1)	-0.0554 [-0.0645, -0.0462]	< 1e-16	3120	supported
H5b_refinement	median Chamfer(method) - median Chamfer(no refinement)	-0.0033 [-0.0046, -0.0021]	< 1e-16	39000	supported
H3	Coarse-label agreement (subsample majority) with literature on preregistered real datasets	7/12 = 58% [28, 85] (exact binomial)	perm. p = 0.125	12	not supported
H4	Controls: matched Gaussians → blob (mean over 17 real datasets); ordinal processes → ringing (not helix)	blob rate 86%; ringing 60% vs helix 40%	-	-	supported

Where the gap comes from. Selecting among the network's own refined candidates with geometric AIC (A5) lowers accuracy to 64.6%, close to B2. Most of the gap to B1 is therefore model selection rather than fitting: with nested families and realistic noise a residual-versus-complexity criterion often prefers the more flexible family, while the classifier has learned the prior over families and noise that generated the data. A5 still chooses only among the classifier's top three candidates, so it is not a purely classical selector. Against B1 the accuracy gap grows with the number of points and is smallest for the smallest clouds; against B2 it is largest for the smallest clouds (Fig. 1). The hardest families are cone, helix and paraboloid. The largest single confusions are helix↔cone, paraboloid↔cone and sphere↔cone, but most helix and paraboloid errors go to other families (Appendix, Fig. 6). Temperature scaling with $T = 0.772$ fitted on the validation stream lowers the expected calibration error from 0.044 to 0.022 and the NLL from 0.397 to 0.373; the NLL-optimal temperature on the test set itself is 0.784 (Appendix, Fig. 6).

6.2 Real activations

Table 4 lists every real dataset and control, and Fig. 3 shows the clouds with the fitted shapes. As preregistered, H3 uses the majority coarse label over random 80% subsamples. It agrees with the literature on 7/12 preregistered datasets; the label-permutation null gives $p = 0.125$, so **H3 is not supported** (it is not falsified either, since agreement is not below 50%). Post hoc, the full-cloud label would agree on 8/12. The twelve datasets count the toy model R3 twice, with 6 and with 8 features; this split was fixed in the code before extraction and is logged as a deviation.

Successes. The toy model of superposition with 6 features (R3) is labelled polytope ($p = 1.00$), with a fitted vertex on every feature direction: three orthogonal antipodal pairs ("digons" in the terminology of Elhage et al., 2022), i.e. an octahedron. Llama's character-count manifold (R4) is labelled ringing ($p = 0.74$), a curve as in

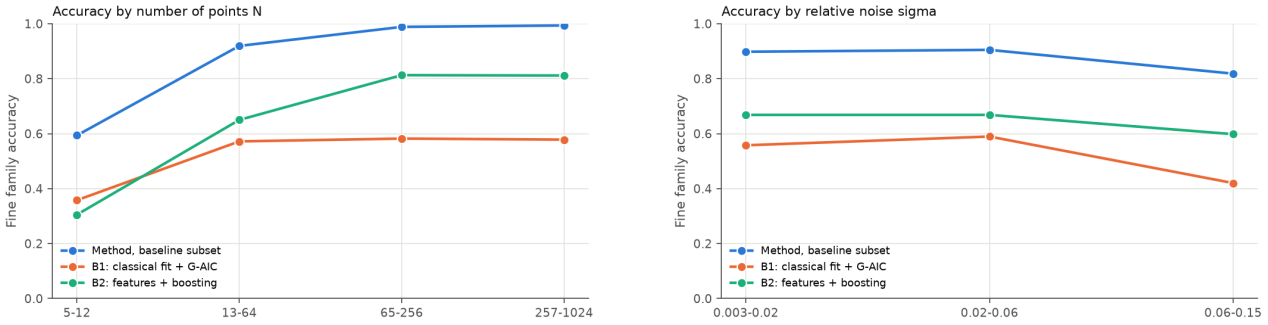


Figure 1: Fine accuracy by number of points and by relative noise on the baseline subset (the same clouds for every line). A3 is omitted because its accuracy equals the method's by construction.

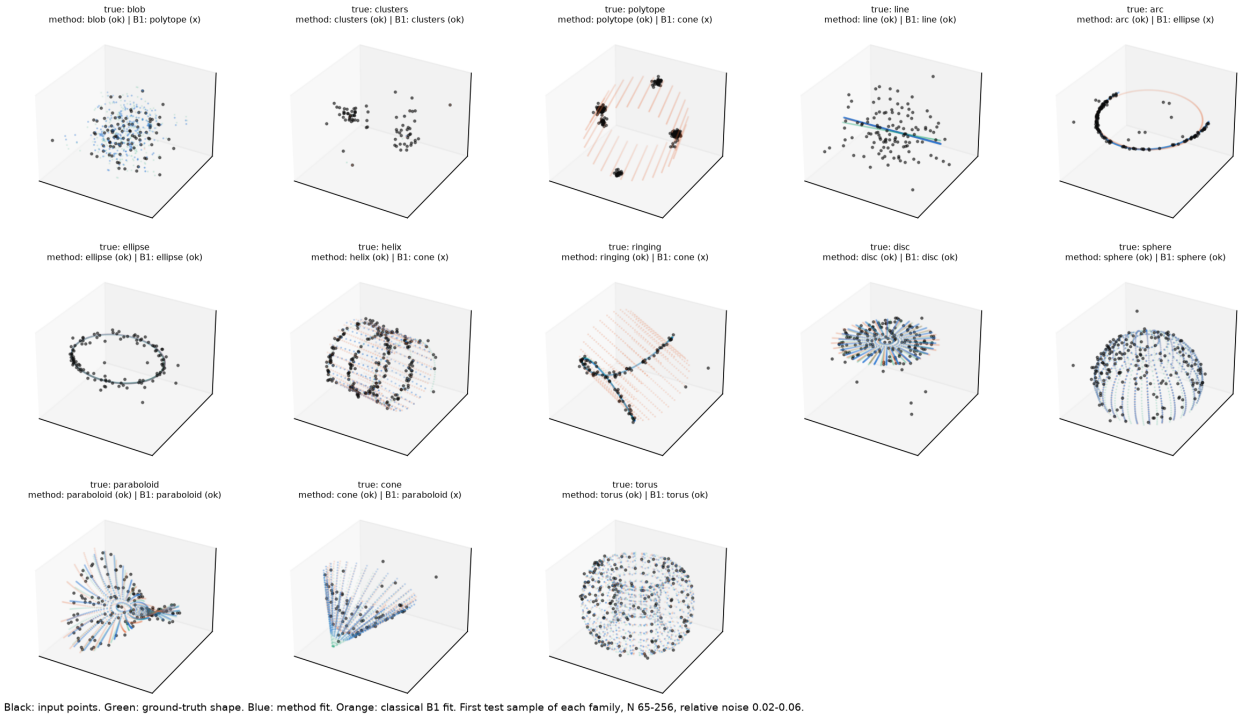


Figure 2: The first synthetic test cloud of every family in the cell with 65–256 points and relative noise 0.02–0.06: input points (black), ground-truth shape (green), method fit (blue), classical B1 fit (orange).

Gurnee et al. [2025]; under the preregistered rule a helix claim is not made because ringing is not rejected. Temperature (R8a) is labelled ringing, the arch PCA produces when similarity decays smoothly with an ordinal variable [Novembre and Stephens, 2008, Antognini and Sohl-Dickstein, 2018], which agrees with the "line" of Bhalla et al. [2026] at the coarse level. Colours (R7) are labelled cone, a curved surface as reported (paraboloid in Bhalla et al., 2026), although the cloud carries the $\lambda_3/\lambda_4 < 1.5$ flag, so its third principal axis is not well separated from the fourth.

Misses and their likely causes (post-hoc inspection, not tested). At the coarse level, H3 is missed on five datasets. (i) GPT-2's positional embeddings (R1) are a 1D spiral whose radius varies along the curve; no family represents it (helix has constant radius, ringing three harmonics), and the classifier selects sphere ($p = 0.65$), a surface the spiral lies on. The held-out residual cannot reveal this error because held-out points also lie on the sphere. (ii) Years (R6) keep only 25% of the variance in the top three components and are flagged ($\lambda_3/\lambda_4 < 1.5$): the 3D view cuts through a higher-dimensional structure. (iii) Age (R8b) is not one smooth curve: ordered by age, consecutive points jump apart around round decade values; it is labelled clusters. (iv) True/false city statements (R9) form two overlapping clouds of very different spread, outside the vocabulary (labelled ringing). (v) The 8-feature toy model (R3) is labelled polytope on the full cloud ($p = 1.00$), but most 80% subsamples, which keep 6 of its 8 vertices, are labelled sphere (56%), so it counts as a miss under the preregistered rule. At the fine level, two more failures do not affect H3: weekdays and months (R5) form one tight cluster per token, 7 and 12 clusters, more than the 6 centroids the clusters family allows, and the fitter threads a helix through them; the grokked embeddings (R2) keep only 32% of the variance in the top three components, as expected for the several Fourier circles of a grokked model [Nanda et al., 2023], and are labelled helix instead of a circle.

Controls (H4). Matched-covariance Gaussians are labelled blob in 86% of draws (mean over the 17 real datasets, exploratory ones included; per dataset the rate falls below the preregistered 80% for the smallest clouds, the toy models and per-concept means, see `reports/deviations.md`), and the ordinal-process controls ringing in 60% versus helix in 40%, so H4 passes under the implemented criterion (the mean blob rate rather than a per-dataset 80% threshold). A post-hoc split shows that this pass relies on the random walks (ringing in 100%): the smooth Gaussian-process controls alone are labelled helix in 80%. The classifier therefore confuses smooth ordinal processes with helices, the confound the preregistered helix rule guards against; for the fully evaluated Gaussian-process control the held-out residual favours ringing and the rule rejects the helix claim. Post hoc, the same rule was applied to every full-cloud helix fit with held-out folds: it rejects the helix claim for `C1_gp_0` but accepts it for `R2_grokking_WE`, `R5a_llama_weekdays`, `R5b_llama_months`, `R5b_llama_months_means`, all of which the literature describes as circles. The rule separates helices from ringing, not from circles or from clusters threaded by a helix. Randomly initialised GPT-2 positional embeddings and grokking embeddings are labelled blob, but randomly initialised Llama is labelled clusters for the month prompts and polytope for the weekday prompts: Fig. 4 shows genuine clusters, one per input token, because a random transformer still separates token identities. The preregistered expectation "blob" for this control was wrong, not the fitter.

7 Discussion and limitations

What the results support. On synthetic clouds drawn from the 13-family generative model, a network trained only on synthetic data identifies the family and recovers the shape better than a classical pipeline that fits every family and selects with geometric AIC (H1, H2), and its probabilities are well calibrated after temperature scaling. This is a *systematic* claim within the synthetic distribution (3120 paired clouds for H1 and H2, 20 per family and cell over all point-count and noise bins), with paired bootstrap CIs. The ablation A5 attributes most of the advantage to model selection: given the same refined candidates, geometric AIC selects much worse than the learned classifier.

What the results do not support. The fitter is not yet a reliable replacement for by-eye shape calls on real activations. Coarse agreement with the literature (H3) is not above a permutation null. Post-hoc inspection suggests causes for the coarse misses: a curve outside the vocabulary (GPT-2 positions), a top-3 projection of higher-dimensional structure (years), clouds unlike any training family (age, true/false statements) and an unstable label on an 8-point cloud. At the fine level, more than six discrete groups (weekdays, months) are

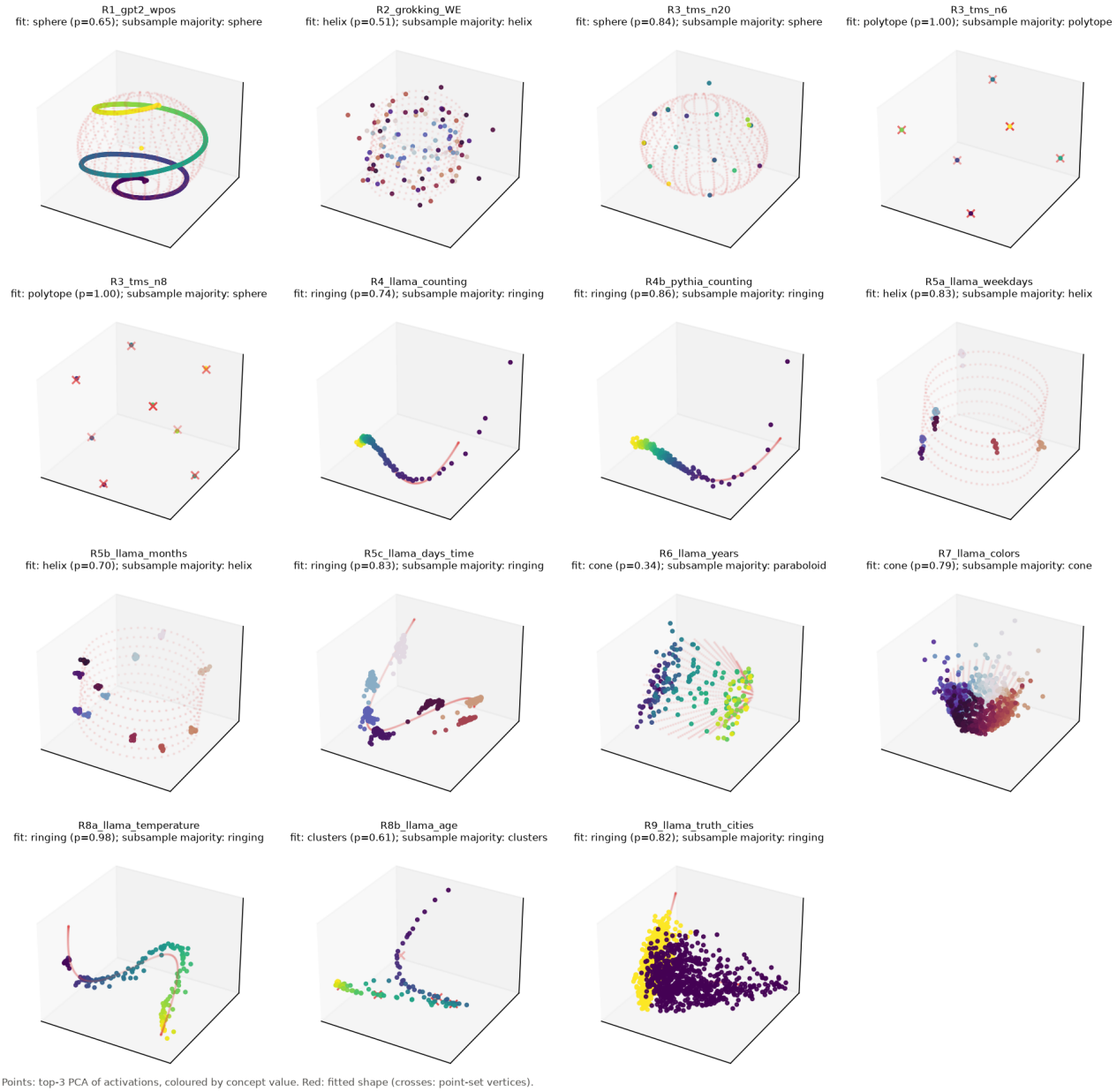


Figure 3: Real activation clouds in their top-3 PCA coordinates, coloured by concept value, with the fitted shape (red; crosses for point-set vertices). Titles give the full-cloud fit with its calibrated probability and the majority label over 200 random 80% subsamples.

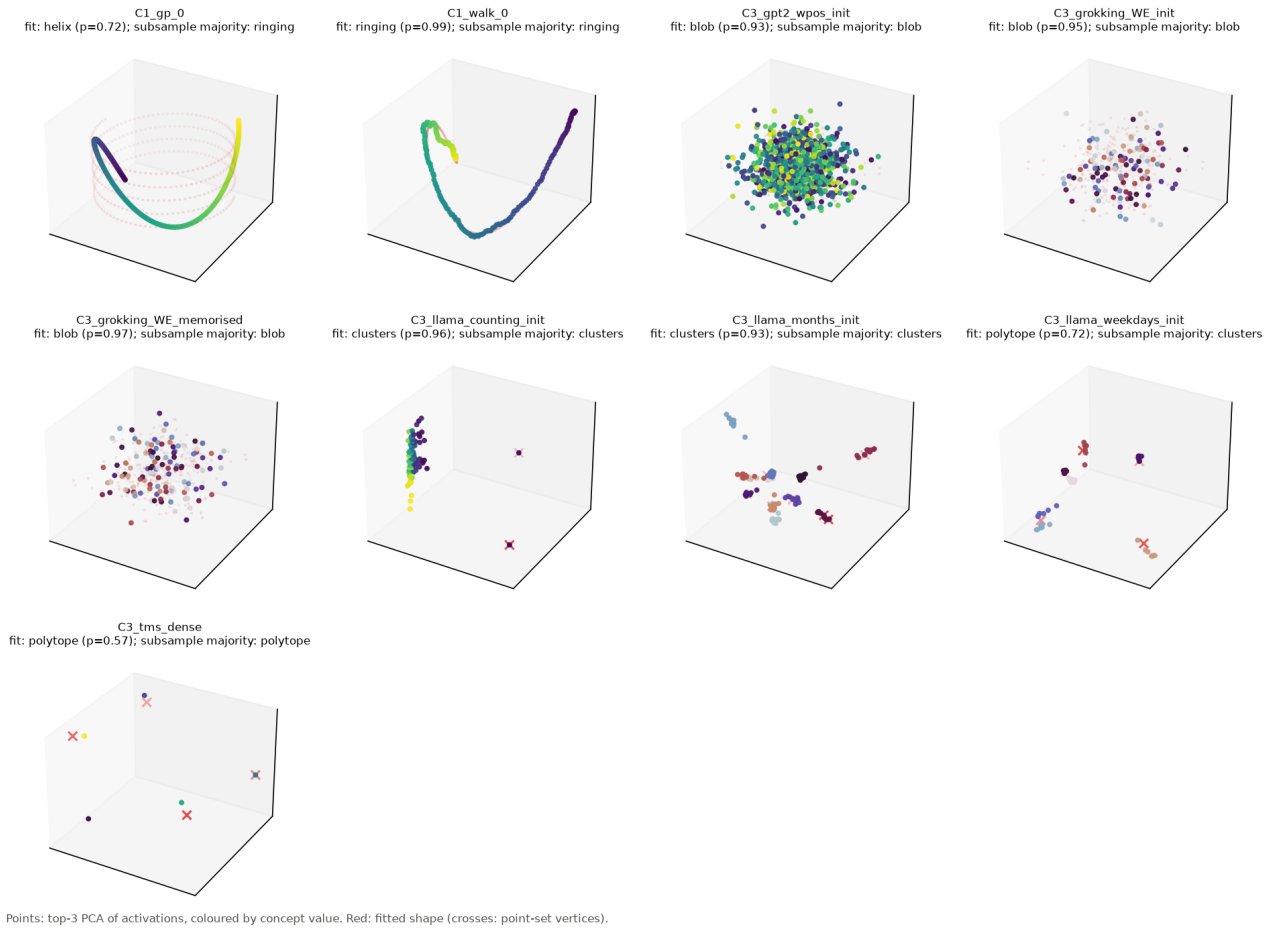


Figure 4: Controls: a smooth Gaussian process and a random walk over an ordinal index (C1), and network controls (C3): randomly initialised GPT-2, Llama and grokking weights, the grokking model at memorisation, and a toy model trained on dense features.

Table 4: Real datasets and controls (C1 shows the first of 10 draws of each process). Flags: $\lambda_3/\lambda_4 < 1.5$ (gap); OOD: residual above the synthetic 95th percentile of correct fits at that N. Held-out residuals, matched-Gaussian rates and model/layer details are in the full table in `reports/exp02_real_activations.md`.

Dataset	**N x D**	**Top-3 var.**	**Expected (coarse / fine)**	**Fit (full cloud)**	**p(fit)**	**Subsample majority (freq.)**	**Flags**
C1_gp_0 (prereg)	200 x 768	99%	1d / ringing	helix	0.72	ringing (58%)	-
C1_walk_0 (prereg)	200 x 768	83%	1d / ringing	ringing	0.99	ringing (100%)	-
C3_gpt2_wpos_init (prereg)	1023 x 768	1%	3d / blob	blob	0.93	blob (100%)	gap<1.5
C3_grokking_WE_init (prereg)	113 x 128	9%	3d / blob	blob	0.95	blob (100%)	gap<1.5
C3_grokking_WE_memorised	113 x 128	15%	- / -	blob	0.97	blob (100%)	gap<1.5
C3_llama_counting_init (prereg)	150 x 4096	14%	3d / blob	clusters	0.96	clusters (98%)	gap<1.5, OOD
C3_llama_months_init (prereg)	144 x 4096	19%	3d / blob	clusters	0.93	clusters (82%)	gap<1.5, OOD
C3_llama_weekdays_init (prereg)	49 x 4096	34%	3d / blob	polytope	0.72	clusters (70%)	gap<1.5, OOD
C3_tms_dense	5 x 3	100%	- / -	polytope	0.57	polytope (100%)	-
R1_gpt2_wpos (prereg)	1023 x 768	91%	1d / helix	sphere	0.65	sphere (74%)	-
R2_grokking_WE (prereg)	113 x 128	32%	1d / arc, ellipse	helix	0.51	helix (43%)	gap<1.5
R3_tms_n20	20 x 3	100%	0d / polytope, clusters	sphere	0.84	sphere (78%)	-
R3_tms_n6 (prereg)	6 x 3	100%	0d / polytope, clusters	polytope	1.00	polytope (100%)	-
R3_tms_n8 (prereg)	8 x 3	100%	0d / polytope, clusters	polytope	1.00	sphere (56%)	-
R4_llama_counting (prereg)	150 x 4096	69%	1d / helix	ringing	0.74	ringing (92%)	-
R4b_pythia_counting	150 x 1024	75%	1d / helix	ringing	0.86	ringing (100%)	-
R5a_llama_weekdays (prereg)	49 x 4096	66%	1d / arc	helix	0.83	helix (58%)	gap<1.5
R5a_llama_weekdays_means	7 x 4096	78%	1d / arc	sphere	0.40	helix (74%)	gap<1.5
R5b_llama_months (prereg)	144 x 4096	55%	1d / arc	helix	0.70	helix (87%)	gap<1.5
R5b_llama_months_means	12 x 4096	60%	1d / arc	helix	0.40	helix (51%)	gap<1.5
R5c_llama_days_time	336 x 4096	58%	1d / arc	ringing	0.83	ringing (86%)	gap<1.5
R6_llama_years (prereg)	199 x 4096	25%	1d / helix	cone	0.34	paraboloid (62%)	gap<1.5
R7_llama_colors (prereg)	900 x 4096	66%	2d / paraboloid	cone	0.79	cone (83%)	gap<1.5
R8a_llama_temperature (prereg)	150 x 4096	90%	1d / line	ringing	0.98	ringing (100%)	-
R8b_llama_age (prereg)	99 x 4096	85%	1d / line	clusters	0.61	clusters (96%)	OOD
R9_llama_truth_cities (prereg)	1496 x 4096	53%	0d / clusters	ringing	0.82	ringing (100%)	OOD

threaded with a helix. These explanations were not tested. In each case the classifier still returns a label, because a closed-set classifier must pick one of its 13 families; its calibrated probability can be low (0.34 for years) or high (1.00 for the 8-point toy model). Smooth ordinal controls are frequently labelled helix (80% of the Gaussian-process controls, post hoc), so a helix label from the classifier alone is not evidence of a helix. The preregistered two-part rule (probability ratio and held-out residual) is necessary but not sufficient: it rejected the helix claim for the one Gaussian-process control on which it was run (the other draws got only the full-cloud fit) but accepted it for `R2_grokking_WE`, `R5a_llama_weekdays`, `R5b_llama_months`, `R5b_llama_months_means`, which the literature describes as circles.

Implications for practice. (1) Report the shape of the top-3 projection together with the explained variance and the λ_3/λ_4 gap; a low gap means the 3D picture is a slice. (2) Treat "helix" as a hypothesis to be tested with held-out data against the PCA ringing artifact [Novembre and Stephens, 2008, Antognini and Sohl-Dickstein, 2018, Shinn, 2023] and against discrete clusters or a circle. (3) Controls with random weights are not structureless when inputs differ in token identity; a token-identity-matched control is the right null for prompt-based datasets. (4) Exclude sequence position 0 (BOS or the first token) when averaging activations: the starting token carries massive activations and attracts attention as a sink [Sun et al., 2024, Xiao et al., 2024], which can dominate class means.

Limitations.

- **One seed, no trained ablations yet.** All results come from one trained model (`main_s0`); the seed, learning-rate and inference-hyperparameter sweeps and the trained ablations A1, A2, A4, A6 (including H5a) are implemented but postponed to later stages (`reports/deviations.md`). Synthetic CIs therefore reflect test-sample variation only, not training variation.
- **Baselines on a subset.** B1 and B2 were run on 20 clouds per family and cell (3120 clouds) because B1 runs up to 61 refinements per cloud; all comparisons with baselines use this subset (H5b is paired on the full test set). B1 is also handicapped in two ways found after the run: its geometric-AIC residual J is measured to the nearest of a finite set of shape samples (a 40×40 grid for surfaces), which adds discretisation error comparable to the lowest noise levels, and in batches containing 5- or 6-point clouds it skips 5- and 6-cluster candidates (`reports/deviations.md`).
- **Closed vocabulary.** 13 families with label margins between nested families; shapes outside the vocabulary are forced into one of them. There is no abstention beyond the OOD and gap flags.
- **Synthetic prior.** Training data follow hand-designed families and nuisances; real activation clouds differ, for example in token-level clusters, and we did not test which difference causes the synthetic-to-real gap.
- **Top-3 PCA only.** All statements concern the top-3 projection, as the task specifies; intrinsic manifolds can differ, and causal roles of these representations are not tested.
- **Literature labels as ground truth.** H3 compares against published descriptions that were themselves made by eye, often with different models, layers or prompts (we use Llama-3.1-8B for all preregistered LLM datasets). The grokked model was trained for fewer epochs than in Nanda et al. [2023], and one weekday prompt differs by one word from Engels et al. [2025].

8 Conclusion

We built a catalog of empirically reported neural geometries, a 13-family shape vocabulary with unique canonical parameters, and a Set Transformer that maps a top-3 PCA point cloud to a family and its parameters, trained only on synthetic shapes with a shape-to-shape Chamfer loss. On held-out synthetic data it identifies the family in 87.9% of clouds and beats a classical fit-every-family pipeline with geometric-AIC selection on both family identification and shape recovery (H1, H2 supported). On real activations, coarse agreement with published shape descriptions (7/12) is not above a permutation null (H3 not supported), and controls show that smooth ordinal processes are often mistaken for helices. The failure modes suggested by post-hoc inspection—out-of-vocabulary curves, many discrete groups, unequal clusters and low-variance projections—point to the next steps: a coverage check that penalises fitted shapes without supporting points, a more flexible Fourier-curve

family, more cluster centroids, a helix test against circles and clusters as well as ringing, and abstention for clouds outside the training distribution.

Reproducibility

All code (MIT license), configurations, results files, the preregistration and the deviation log are at https://github.com/BurnyCoder/geometrical-shape-fitting-in-llm-mechanistic-interpretability. `uv run python run.py report --config main.yaml` regenerates every number, table and figure in this paper from the results files, except the catalog table (rebuilt from the catalog by the `catalog` phase) and the synthetic gallery figure (the `gallery` phase re-fits its 13 clouds with the trained checkpoint). The trained fitter and the run logs (training, evaluation, and activation extraction with the LLM prompts; the counting corpus is logged only as the first 200 characters per document, and the exploratory Pythia run logs no prompts) are on the Hugging Face Hub at https://huggingface.co/BurnyCoder/neurogeom-shape-fitter-main-s0 (collection "Fitting geometric shapes to neural geometry"), and an interactive explorer of all fits is at https://burnycoder.github.io/geometrical-shape-fitting-in-llm-mechanistic-interpretability/.

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A Catalog of empirically reported shapes

Shape	Concept	Model / layer	Top-3 PCA	Family	Source
circle (overall a cone; PC1 = intensity/radius)	days of the week, months of the year (also 20th-century years in GPT-2)	GPT-2-small L7, Mistral-7B L8 (SAE feature clusters); Mistral-7B L30, Llama-3-8B L5 (PCA of hidden states on modular-arithmetic prompts)	yes (PCs 1-2 or 2-3)	arc	Engels et al. [2025]
circle ("days of the week organized in a cycle")	days of the week (prompt "It's {time} on day")	Llama-3.1-8B L19, last token	yes (Fig. 2 left; Fig. 8)	arc	Bhalla et al. [2026]
circles (one per key Fourier frequency; "clock" in Zhong et al.'s terms)	residues in modular addition (a+b mod p)	1-layer transformer trained from scratch, p=113 (Nanda et al.) / p=59 (Zhong et al.)	partial (each key frequency is a circle in its own plane; Nanda et al. find 6 frequencies in W_E, so the top-3 PCs mix circles)	arc	Nanda et al. [2023], Zhong et al. [2023]
ring / square grid / hexagonal lattice (in-context graph structure)	graph defined in context by random walks over tokens	Llama-3.1-8B, after layer 26	yes (top-2 PCs; with a days-of-week prior the ring is in PCs 3-4)	arc	Park et al. [2025a]
approximately circular arrangement (valence-arousal circumplex; the authors note the method does not guarantee it)	emotions (27 GoEmotions categories)	Llama-3.1-8B-Instruct, Qwen3-8B, Qwen3-14B	no (PCA of steering vectors, not of raw activations)	arc	Sun et al. [2026]
ring	head direction (neuroscience analogue)	mouse anterodorsal thalamic nucleus	no (shown with Isomap; the paper notes PCA often fails to reveal the ring)	arc	Chaudhuri et al. [2019]
helix	learned absolute positional embeddings (W_pos, 1024 x 768; row 0 is an outlier and dropped)	GPT-2 small/medium/large and the Stanford GPT-2 models (weak in GPT-2 XL)	yes (˜90% variance in 3 PCs for GPT-2 small)	helix	Yedidia [2023]
twisting helix-like rippled curve (1D manifold; the top 6 PCs hold 95% of the variance)	character count since last line break	Claude 3.5 Haiku L2; replicated in Llama-3.1-8B L5 (PC1-3: 97.1% of variance) and Qwen3-8B L5 (PC1-3: 95.3%) by Sinii & Balagansky 2026	yes	helix	Gurnee et al. [2025], Sinii and Balagansky [2026]
generalized helix (line + circles with periods 2, 5, 10, 100)	integers 0-99 (helix fit); 0-360 only for the layer-0 Fourier analysis of GPT-J	GPT-J (main; Fourier analysis of the residual after layer 0, helix fits at every layer); Pythia-6.9B and Llama-3.1-8B (appendix replication)	partial (PC1 ˜ linear; helix only in fitted subspace)	helix	Kantamneni and Tegmark [2025]

Shape	Concept	Model / layer	Top-3 PCA	Family	Source
helix	years (prompt "The date is year")	Llama-3.1-8B L19, last token	unknown (the Fig. 2 caption lists age, temperature, day and color as its PCA examples; Fig. 7 shows years as SAE tuning curves)	helix	Bhalla et al. [2026]
Fourier features with periods 2, 2.5, 5 and 10 (a cos/sin pair implies a circle, but no circle is plotted)	numbers in addition	GPT-2-XL fine-tuned on addition (main); GPT-J and Phi-2 with few-shot prompts	unknown	arc	Zhou et al. [2024]
line	temperature ("Today it's {f} degrees Fahrenheit outside"), age ("They are {age} years old."), formality, sentence length	Llama-3.1-8B L19, last token	yes for temperature and age (Fig. 2 left)	line	Bhalla et al. [2026]
line / linear map (latitude, longitude, year)	places and historical entities	Llama-2 7B/13B/70B, mid-depth (and Pythia in an early experiment)	not reported (linear probes, no top-3 PCA plot)	line	Gurnee and Tegmark [2024]
sphere (cities); circle (dates, time of day); semicircle (events 1900-2000); log-line (durations); clusters (seasons)	time and space concepts	Qwen2.5-3B, Llama-3.2-3B, gemma-2-2b (main); Llama-3.1-8B/70B (scale check)	no (paper: PCA misses subspaces not aligned with max variance)	sphere	Tiblias et al. [2026]
sphere (sky positions of stars and constellations on S^2)	celestial objects	Qwen3-32B, Qwen3-235B, gpt-oss-120b, Llama-3.3, Mixtral-8x22B, Mistral Large 2 (main example, L66), GLM-4.5-Air	no (rises to about the top-8 PCs, mostly PCs 2-4; never the top 3)	sphere	Berdnikov and Liokumovich [2026]
paraboloid ("circular hue and lightness dimensions")	colors (prompt "The hex code {h_code} is for the color")	Llama-3.1-8B L19, last token	yes (Fig. 2 left)	paraboloid	Bhalla et al. [2026]
disc ("pizza") or torus in the first layer; circle in later layers	modular addition	small MLPs (MLP-Add, MLP-Concat) and attention models trained from scratch	partial (disc in PCs 1-2 with >99% of the variance; the MLP-Concat torus is a flat torus spread over 4 PCs of about 25% each)	torus	Moisescu-Pareja et al. [2026]
filled disc ("pizza") of averaged pair embeddings (E_a+E_b)/2 inside the circle of token embeddings	pairs of residues in modular addition (pizza algorithm, used when attention is near-uniform)	1-layer transformers trained from scratch, p=59	yes (embedding circle in PCs 1-2, sometimes other PC pairs; the pair representations fill its interior)	disc	Zhong et al. [2023]
torus	grid-cell modules (neuroscience analogue)	rat medial entorhinal cortex	no (shown after the top 6 PCs are reduced to 3D with nonlinear UMAP)	torus	Gardner et al. [2022]
narrow cone (anisotropy; a direction-concentration statement in high dimensions, not a fitted cone surface)	all contextual embeddings	GPT-2, BERT, ELMo	no PCA view (the cone is inferred from the mean cosine similarity of random words)	cone	Ethayarajh [2019]
true and false statements separate (linearly) in the top-2 PCs	true vs false statements (e.g. cities, 1,496 statements)	LLaMA-2 7B/13B/70B (LLaMA-2-13B layer 15; layer 12 for cities+neg_cities)	yes	clusters	Marks and Tegmark [2024]
uniform polytopes - digons (antipodal pairs), triangles, pentagons, tetrahedra, square antiprisms; an octahedron of antipodal pairs in the correlated-features experiment	features in superposition	toy ReLU autoencoders	partial (direct only for 2-3 hidden dimensions; in larger models the polytopes sit in orthogonal subspaces)	polytope	Elhage et al. [2022]
simplices / polytopes; hierarchy orthogonal	WordNet categories	Gemma-2B (main), LLaMA-3-8B (appendix)	no	polytope	Park et al. [2025b]
parallelogram/trapezoid "crystals", lobes	word-pair function-vector differences (crystals; the search among SAE features "found mostly noise"); SAE feature co-occurrence (lobes)	Gemma-2-2b (layer-0 residual activations for crystals; layer-12 residual SAE for lobes)	no (PC1 of the layer-0 differences mainly encodes word length; crystals appear only after LDA)	polytope	Li et al. [2025]

Shape	Concept	Model / layer	Top-3 PCA	Family	Source
hierarchical tree	geographical coordinates → country ("The geographical coordinates {lat, lon} are in the country of")	Llama-3.1-8B L19	unknown (not plotted)	out-of-vocab	Bhalla et al. [2026]
fractal inside a 2-simplex	belief states of a hidden Markov process (Mess3)	4-layer transformer, d_model=64	not reported (no PCA in the paper)	out-of-vocab	Shai et al. [2024]
Klein bottle	3x3 high-contrast natural image patches	natural image patches (van Hateren database); not CNNs (for CNN filters see cnn_filter_circle)	no (needs 4D)	out-of-vocab	Carlsson et al. [2008]
circle of rotating edge filters	spatial convolution filters, recurring "among all networks, depths, and during training" (follow-up work finds the three-circle subset of the Klein bottle in first-layer CIFAR-10 filters)	CNNs on MNIST, CIFAR-10, SVHN, ImageNet	n/a	arc	Brüel Gabrielsson and Carlsson [2018]
nested "onion" shells (magnitude-coded position)	token identity at each sequence position, with the position coded by magnitude (repeat task)	small GRUs (larger GRUs learn linear representations instead)	not reported (no PCA in the paper)	out-of-vocab	Csordás et al. [2024]

B Additional results

B.1 Unambiguous synthetic subset

Synthetic clouds that a simpler family also fits equally well (judged from B1's per-family fits) are tagged ambiguous (34.0% of the baseline subset). Table 6 repeats Table 2 without them.

Table 6: Systems on the baseline subset without ambiguous clouds.

System	n	Fine accuracy [95% CI]	Macro-F1 [95% CI]	Coarse accuracy	Superset-OK accuracy	Median Chamfer to GT [95% CI]	Mean residual	Median axis error (deg)
Method (network + refinement)	2060	94.0% [93.0, 95.0]	0.922 [0.908, 0.935]	96.4%	94.3%	0.119 [0.111, 0.127]	0.091	0.7
A3: no refinement	2060	94.0% [93.0, 95.0]	0.922 [0.908, 0.935]	96.4%	94.3%	0.127 [0.119, 0.132]	0.101	1.1
A5: G-AIC selection	2060	72.0% [70.1, 74.0]	0.654 [0.637, 0.671]	81.5%	76.5%	0.138 [0.134, 0.144]	0.097	0.6
A5: hybrid selection	2060	92.6% [91.4, 93.7]	0.902 [0.888, 0.916]	94.8%	93.1%	0.119 [0.111, 0.128]	0.090	0.7
B1: classical fit + G-AIC	2060	69.7% [67.6, 71.6]	0.636 [0.615, 0.654]	82.5%	75.4%	0.154 [0.147, 0.160]	0.103	0.6
B2: features + boosting	2060	72.3% [70.3, 74.2]	0.697 [0.674, 0.715]	92.5%	76.2%	0.152 [0.147, 0.159]	0.099	0.6
B3a: always blob	2060	4.5% [3.6, 5.4]	0.007 [0.005, 0.008]	4.5%	4.5%	0.523 [0.517, 0.529]	0.119	10.2
B3b: always arc (B1 arc fit)	2060	7.3% [6.2, 8.5]	0.011 [0.009, 0.012]	36.3%	7.3%	0.528 [0.516, 0.538]	0.273	0.3
B3c: random shape	2060	8.3% [7.2, 9.5]	0.079 [0.068, 0.090]	34.1%	11.6%	1.114 [1.092, 1.138]	0.582	58.6

B.2 Training, confusion and calibration

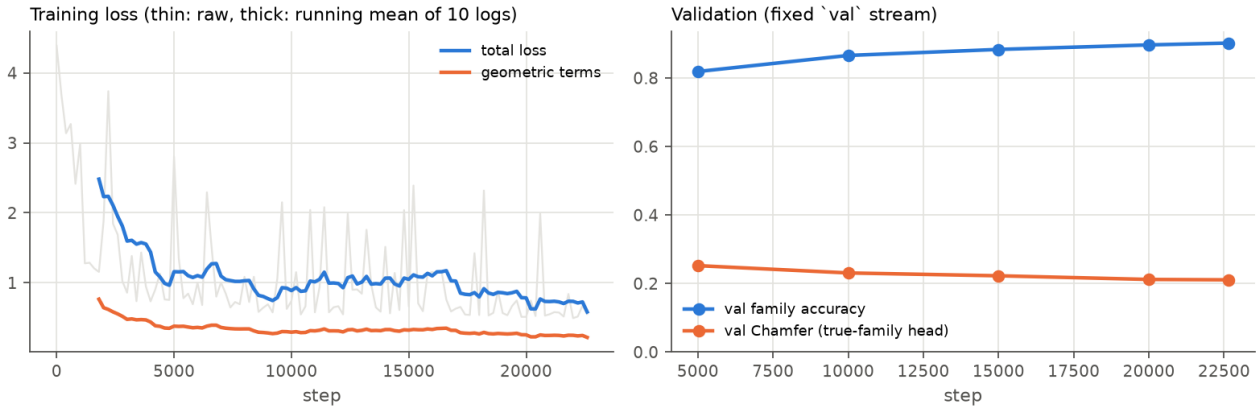


Figure 5: Training loss (left) and validation family accuracy and teacher-forced Chamfer distance on the fixed validation stream (right). Steps 0–9999 of the loss curve were recovered from the log of the crashed first attempt (`reports/deviations.md`).

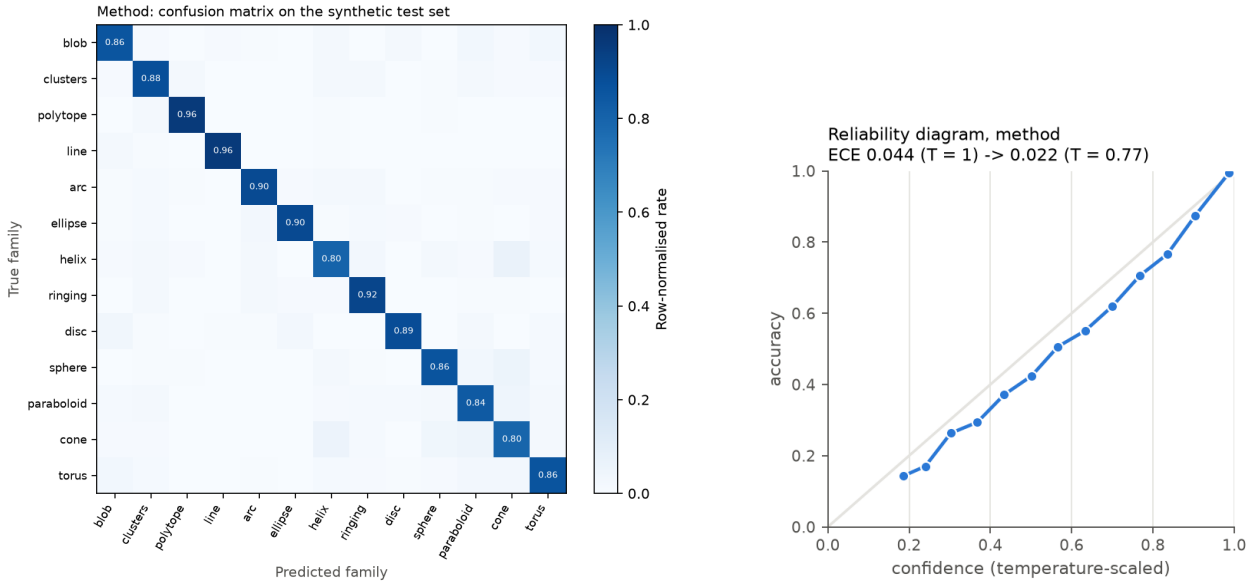


Figure 6: Left: confusion matrix of the method on the full synthetic test set (row-normalised). Right: reliability diagram after temperature scaling.

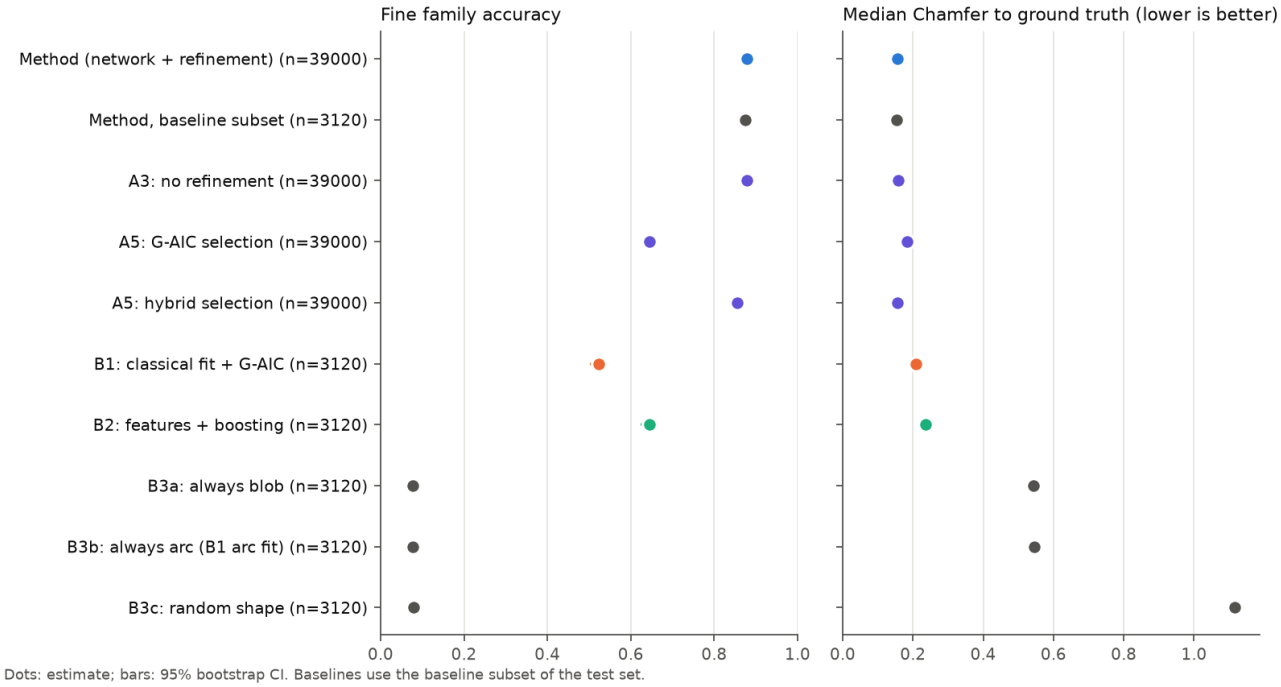


Figure 7: Fine accuracy and median Chamfer distance of all systems with 95% bootstrap CIs.

B.3 Models

Table 7: Trained fitters.

Run	Encoder	Training objective	Data / curriculum	Params	Steps	Minutes	Val. accuracy	Test fine accuracy	Test median Chamfer
main_s0	Set Transformer	CE + Chamfer + residual + aux	full synthetic mix	1,712,093	22643	110	90.2%	87.9%	0.156

AI statement

This codebase and paper were written by Claude Code (Anthropic), an AI coding agent, which executed a specification provided by the human author; the human author provided the research question, constraints and standards to follow. Human cleanup and rewriting are in progress.